WHO MOVED MY HEAP · 04 SOLUTIONS

Solutions

Tools, runtime options and code that move less heap

Intro | Problem | Definitions | **Solutions** | AI skills | Wrap-up

04

Solutions at a glance

Solution	Use it to	Measured here
JDK CLI tooling	See what is on the heap, and what a workload drops	Measure first
Java Object Layout	Size one object or a whole collection	`HashMap$Node` 32 B to 24 B
Java Microbenchmark Harness	Count bytes and objects per call	176 B per `anyMatch`
Object pooling	Reuse costly objects, not small ones	`Integer` -128 to 127
Compact object headers	Shrink every header by 4 B	14% off a `HashMap`
Non-capturing lambdas	Share one lambda instance	32 B to 16 B per call (EA off)
Eclipse Collections	Primitives, and loops without a pipeline	48 B to 8 B per element; 0 B per call